

MAWLANA BHASHANI SCIENCE AND TECHNOLOGY UNIVERSITY

SANTOSH, TANGAIL-1902



DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

Lab Report

Lab Report No : 05

Lab Report on : Development and Deployment of a Responsive Personal Portfolio Using
HTML and Bootstrap 5

Course Title : Web Application Development Lab

Course Code : ICT-3204

Submitted By	Submitted To
Name: Kuldip Saha Mugdha ID: IT-22018 3 rd Year, 2 nd Semester Session: 2021-2022 Dept. of ICT, MBSTU	Mr. Md. Anowar Kabir Lecturer DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY MAWLANA BHASHANI SCIENCE AND TECHNOLOGY UNIVERSITY

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Introduction:

In the digital era, having an online presence is essential for personal and professional growth. This lab report documents the process of designing and publishing a personal portfolio. Through this project, I explored web design basics, content organization, and digital presentation—all without the need for advanced coding skills.

Objectives:

The objective of this lab experiment is to design, develop, and deploy a responsive personal portfolio website using modern web development technologies. The website demonstrates the use of **HTML5**, **Bootstrap 5**, **Google Fonts**, **AOS (Animate On Scroll)** animations, and other supporting technologies to create a visually appealing, responsive, and user-friendly portfolio suitable for professional use.

Tools and Technologies Used

- **HTML5** – For structuring the web content
- **CSS3** – For additional styling and layout customization
- **Bootstrap 5** – For responsive design and pre-built UI components
- **Google Fonts** – For enhanced typography
- **AOS (Animate On Scroll) Library** – For scroll-based animations
- **JavaScript** – For interactive features and animations
- **VS Code** – Code editor used for development
- **Web Browser** – Google Chrome for testing and debugging

Development Environment:

- **Code editor:** VS Code
- **Browser:** Chrome / Firefox / Edge (latest)
- Git installed locally

Working Principle:

The working principles of the responsive website are based on the following core ideas:

Bootstrap Responsive Grid

A 12-column grid adjusts layout automatically on mobile, tablet, and desktop using breakpoints.

Mobile-First Design

Design starts from small screens and scales upward for better performance and usability.

Google Fonts Integration

Fonts load from Google's CDN, ensuring consistent typography and faster rendering.

AOS (Animate On Scroll)

Elements animate when they enter the viewport using data-aos attributes and AOS JavaScript.

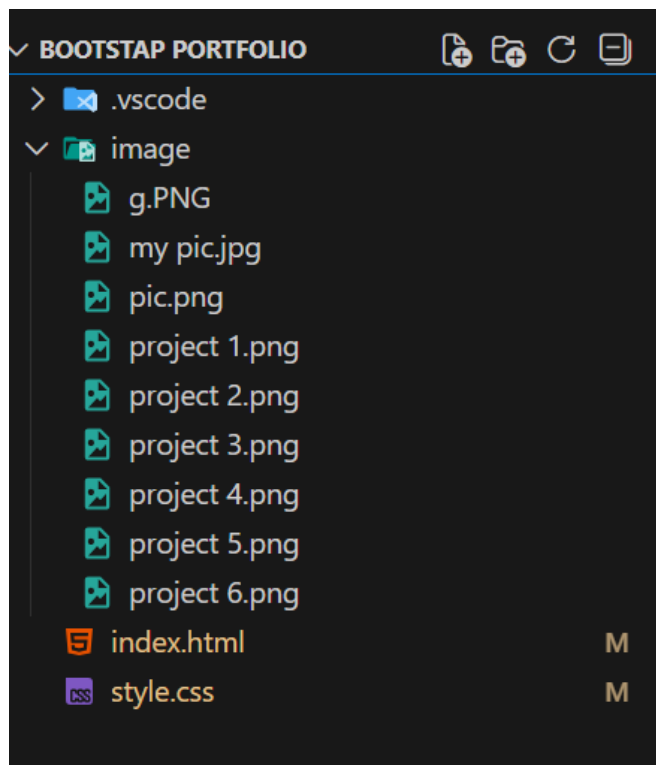
Bootstrap Components

Interactive components like navbar toggler work using Bootstrap's JavaScript bundle.

GitHub Pages Hosting

GitHub Pages directly serves static files (HTML, CSS, JS) from the repository's main branch as a website.

Project Structure:



Important Implementation Notes:

HTML <head> (essential tags and CDN links):

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Mugdha Saha | Portfolio</title> "Mugdha": Unknown word.
  <link rel="stylesheet" href="style.css">
  <link href="https://unpkg.com/aos@2.3.1/dist/aos.css" rel="stylesheet">
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.0.2/dist/css/bootstrap.min.css" rel="stylesheet"
    integrity="sha384-EVSTQN3/azprG1Anm3QDgpJLIm9Nao0Yz1ztcQTwFspd3yD65VohhpuuCOMLAsjC" crossorigin="anonymous">
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Space+Grotesk:wght@400;500;600;700&family=Poppins:wght@300;400;500;600;700" rel="stylesheet">
  <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/font-awesome@6.5.2/css/all.min.css" integrity="sha384-Jc6p6NoJa6st7F1bD7KwTm8BDyb90IAdY1vz9L340ND0t8Dd889u1s009035" crossorigin="anonymous">
</head>
```

Example Responsive Navbar (Bootstrap):

```
<nav class="navbar navbar-expand-lg navbar-dark glass-nav fixed-top">
  <div class="container">
    <a class="navbar-brand fs-3 fw-bold" href="#home">Mugdha Saha</a> "Mugdha": Unknown word.
    <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="#navbarSupportedContent"
      <span class="navbar-toggler-icon"></span>
    </button>
    <div class="collapse navbar-collapse" id="navbarSupportedContent">
      <ul class="navbar-nav ms-auto mb-2 mb-lg-0 gap-2">
        <li class="nav-item">
          <a class="nav-link" href="#home">Home</a>
        </li>
        <li class="nav-item">
          <a class="nav-link" aria-current="page" href="#about">About</a>
        </li>
        <li class="nav-item">
          <a class="nav-link" href="#skills">Skills</a>
        </li>
        <li class="nav-item">
          <a class="nav-link" href="#projects">Projects</a>
        </li>
        <li class="nav-item">
          <a class="nav-link" href="#contact">Contact</a>
        </li>
      </ul>
    </div>
  </div>
</nav>
```

1. Creating the Navigation Bar :

Organized the site using a navigation bar with the following sections:

Home, About, Skills, Projects, Contact.

2. Developing the “Home” Section:

In this section, I introduced myself, described my academic background, and highlighted my technical skills.

Code:

```
<section class="hero" id="home">
  <div class="container">
    <div class="row align-items-center">
      <div class="col-lg-6" data-aos="fade-right">
        <p class="text-uppercase text-light mb-2">Hello, I'm Mugdha</p>
        <h1 class="display-4 fw-bold text-white mb-3">IT student, problem solver, and tech explorer.</h1>
        <p class="text-white-50 mb-4">From "Hello World" in C to IoT prototypes and secure systems, I love learning
        <div class="d-flex flex-wrap gap-3">
          <a class="btn btn-primary btn-lg" href="#projects">View Work</a>
          <a class="btn btn-outline-light btn-lg" href="#about">About Me</a>
        </div>
        <div class="social mt-4 d-flex gap-3">
          <a class="text-white-50 fs-4" href="mailto:mugdha.saha67@gmail.com" title="Email"><i class="fa-solid fa-envelope"></i></a>
          <a class="text-white-50 fs-4" href="tel:+8801840159906" title="Call"><i class="fa-solid fa-phone"></i></a>
          <a class="text-white-50 fs-4" href="#" title="Discord"><i class="fa-brands fa-discord"></i></a>
        </div>
      </div>
      <div class="col-lg-6 text-center" data-aos="fade-left">
        <div class="hero-card shadow-lg">
          
          <div class="d-flex justify-content-between align-items-center mt-3">
            <div>
              <p class="text-white-50 mb-1">Focus</p>
              <h4 class="text-white">Cybersecurity</h4>
            </div>
            <div>
              <p class="text-white-50 mb-1">Building</p>
            </div>
          </div>
        </div>
      </div>
    </div>
  </div>
```

➤ Visual Overview of the “Home” Section:

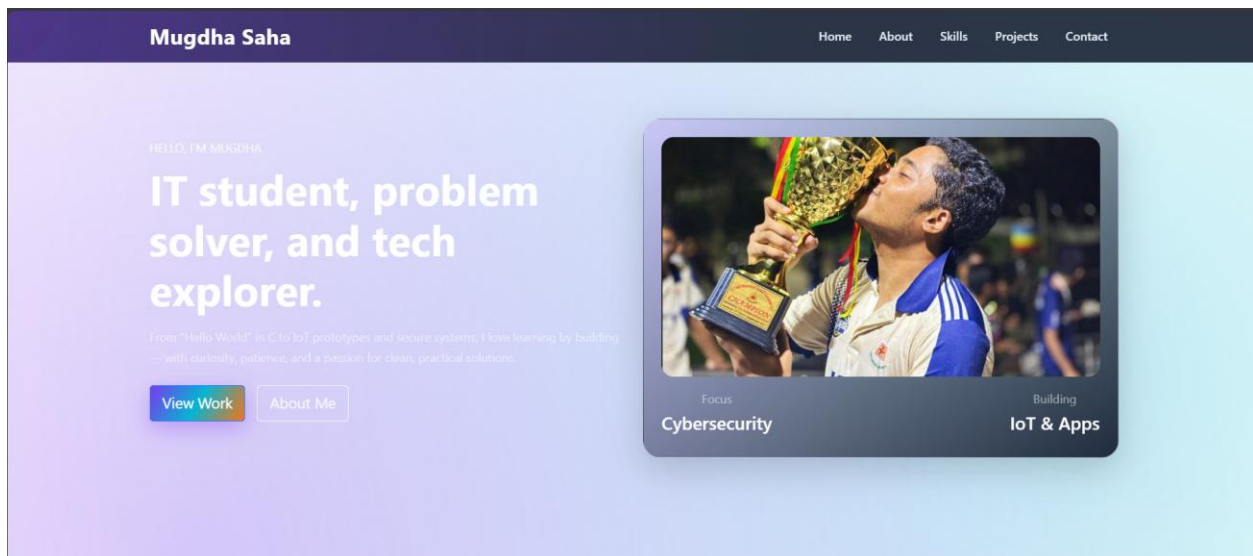


Fig 01: Home Section

2. Developing the “About” Section:

Code:

```
<section class="about my-5" id="about">
  <div class="container">
    <h1 class="text-center section-title" data-aos="zoom-in" data-aos-offset="300" data-aos-duration="500">About

    <div class="row align-items-center g-4">
      <div class="col-lg-6" data-aos="fade-right">
        <p class="lead text-muted">Enthusiastic IT student who learns by building — from desktop apps to IoT prot
        <div class="d-flex flex-wrap gap-3 mt-3">
          <div class="stat-card">
            <h3 class="mb-0">Cyber & IoT</h3>
            <small class="text-white-50">Current deep dives</small>
          </div>
          <div class="stat-card">
            <h3 class="mb-0">CTF</h3>
            <small class="text-white-50">Puzzle-driven learning</small>
          </div>
          <div class="stat-card">
            <h3 class="mb-0">IoT</h3>
            <small class="text-white-50">Physical ↔ digital builds</small>
          </div>
        </div>
      </div>
      <div class="col-lg-6" data-aos="fade-left">
        <div class="glass p-4 rounded-4 shadow-lg">
```

➤ Visual Overview of the “About” Section:

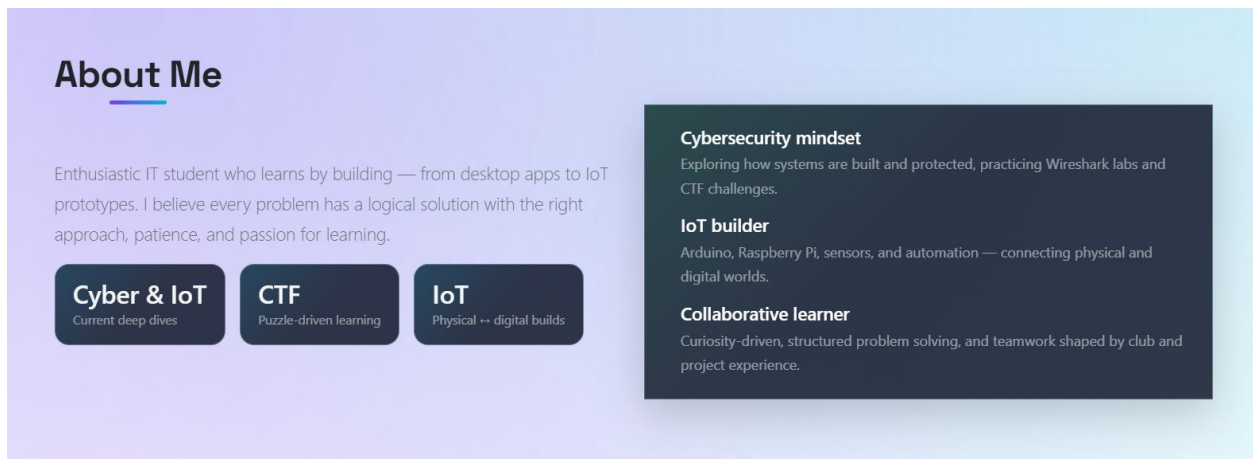


Fig 02: About Me Section

3. Developing the “Education” Section:

Code:

```
<section class="my-5" id="education">
  <div class="container">
    <h1 class="text-center section-title" data-aos="zoom-in">Education</h1>
    <div class="row g-4">
      <div class="col-md-4" data-aos="fade-up">
        <div class="glass p-4 rounded-4 h-100">
          <h5 class="text-white mb-1">Mawlana Bhashani Science and Technology University</h5> "Mawlana": Unkn
          <p class="text-white-50 mb-1">BSc (Engg) in ICT • 2023 – Present</p> "Engg": Unknown word.
          <small class="text-primary">Tangail, Bangladesh</small> "Tangail": Unknown word.
          <p class="text-white-50 mt-2 mb-0">Building foundations in programming, data structures, networking, D
        </div>
      </div>
      <div class="col-md-4" data-aos="fade-up" data-aos-delay="100">
        <div class="glass p-4 rounded-4 h-100">
          <h5 class="text-white mb-1">St. Gregory's High School & College</h5>
          <p class="text-white-50 mb-1">Higher Secondary (Science) • 2019 – 2022</p>
          <small class="text-primary">Dhaka Education Board • GPA 5.00</small>
          <p class="text-white-50 mt-2 mb-0">Strengthened curiosity in tech, math, problem-solving, and teamwork
        </div>
      </div>
      <div class="col-md-4" data-aos="fade-up" data-aos-delay="200">
        <div class="glass p-4 rounded-4 h-100">
          <h5 class="text-white mb-1">Narinda Government High School</h5> "Narinda": Unknown word.
          <p class="text-white-50 mb-1">Secondary (Science) • 2008 – 2019</p>
          <small class="text-primary">Dhaka Education Board • GPA 4.89</small>
          <p class="text-white-50 mt-2 mb-0">Sparked interest in computers, science experiments, and leadership
```

➤ Visual Overview of the “Education” Section:

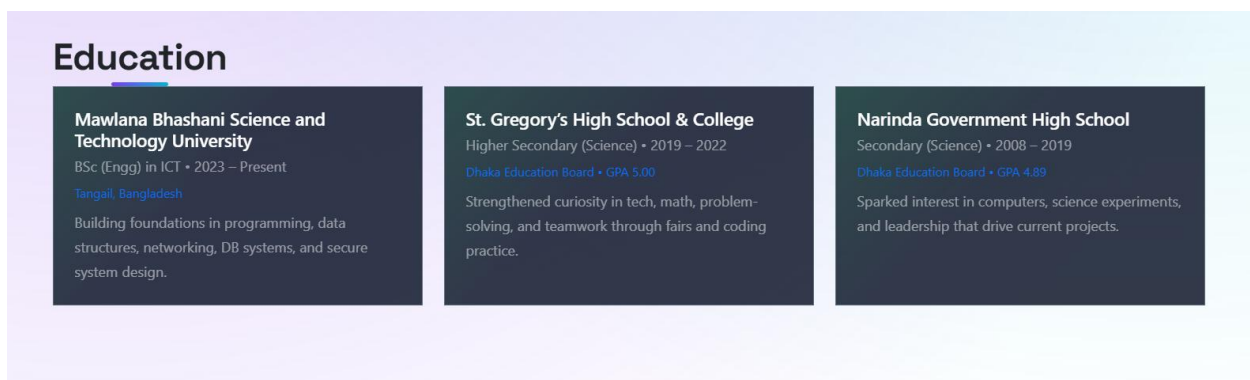


Fig 03: Education Section

4. Developing the “Skills” Section:

Code:

```
<section class="skills my-5" id="skills">
  <div class="container">
    <h1 class="text-center section-title" data-aos="zoom-in">Skills</h1>
    <div class="row g-4">
      <div class="col-lg-4" data-aos="fade-up">
        <div class="glass h-100 p-4 rounded-4">
          <div class="d-flex justify-content-between align-items-center mb-3">
            <h4 class="text-white mb-0">Programming</h4>
            <span class="badge bg-primary">Actively building</span>
          </div>
          <p class="text-white-50">C, C++, Java (Swing), Python scripting, Kotlin basics</p>
          <div class="progress mb-2"><div class="progress-bar bg-primary" style="width: 85%"></div></div>
          <small class="text-white-50">Logic-first problem solving</small>
        </div>
      </div>
      <div class="col-lg-4" data-aos="fade-up" data-aos-delay="100">
        <div class="glass h-100 p-4 rounded-4">
          <div class="d-flex justify-content-between align-items-center mb-3">
            <h4 class="text-white mb-0">Cybersecurity</h4>
            <span class="badge bg-info text-dark">Learning</span>
          </div>
          <p class="text-white-50">Network analysis, Wireshark labs, ethical hacking basics, CTF practice</p>
          <div class="progress mb-2"><div class="progress-bar bg-info" style="width: 70%"></div></div>
          <small class="text-white-50">Build + defend mindset</small>
        </div>
      </div>
    </div>
  </div>
</section>
```

➤ Visual Overview of the “Skills” Section:

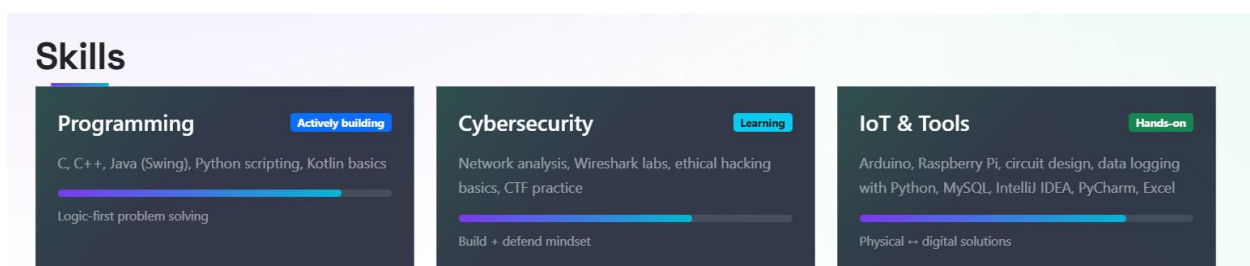


Fig 04: Skills Sections

5. Developing the “Projects” Section:

Code:

```
<section class="projects my-5" id="projects">
  <div class="container">
    <h1 class="text-center section-title data-aos="zoom-in">Projects</h1>
    <div class="row g-4">

      <div class="col-md-4" data-aos="zoom-in">
        <div class="project-card rounded-4 overflow-hidden shadow-lg">
          
          <div class="p-3">
            <h5 class="text-white mb-1">🔥 ICT Premier League Console Game</h5>
            <p class="text-white-50 mb-2">Interactive cricket simulation in C++ with team selection, virtual toss, and 5-over matches. Features randomized events, extras, and real-time scoreboard.</p>
            <div class="d-flex flex-wrap gap-1 mb-2">
              <span class="badge bg-primary">C++</span>
              <span class="badge bg-info text-dark">Console App</span>
              <span class="badge bg-success">Game Development</span>
            </div>
            <a href="https://github.com/Mugdha67/MBSTU-ICT-Batch-19/tree/main/Projects/ICTPL%20Cricket%20Tournament" class="text-white">View Project</a>
          </div>
        </div>
      </div>

      <div class="col-md-4" data-aos="zoom-in" data-aos-delay="100">
        <div class="project-card rounded-4 overflow-hidden shadow-lg">
          
          <div class="p-3">
            <h5 class="text-white mb-1">📊 Simple Calculator (Java)</h5>
            <p class="text-white-50 mb-2">Console-based Java application performing arithmetic operations including addition, subtraction, multiplication, division, and power calculation with a menu-driven interface.</p>
            <div class="d-flex flex-wrap gap-1 mb-2">
              <span class="badge bg-primary">Java</span>
              <span class="badge bg-info text-dark">Console App</span>
              <span class="badge bg-success">2024</span>
            </div>
            <a href="#" class="text-white">View Project</a>
          </div>
        </div>
      </div>

    </div>
  </div>
</section>
```

➤ Visual Overview of the “Projects” Section:

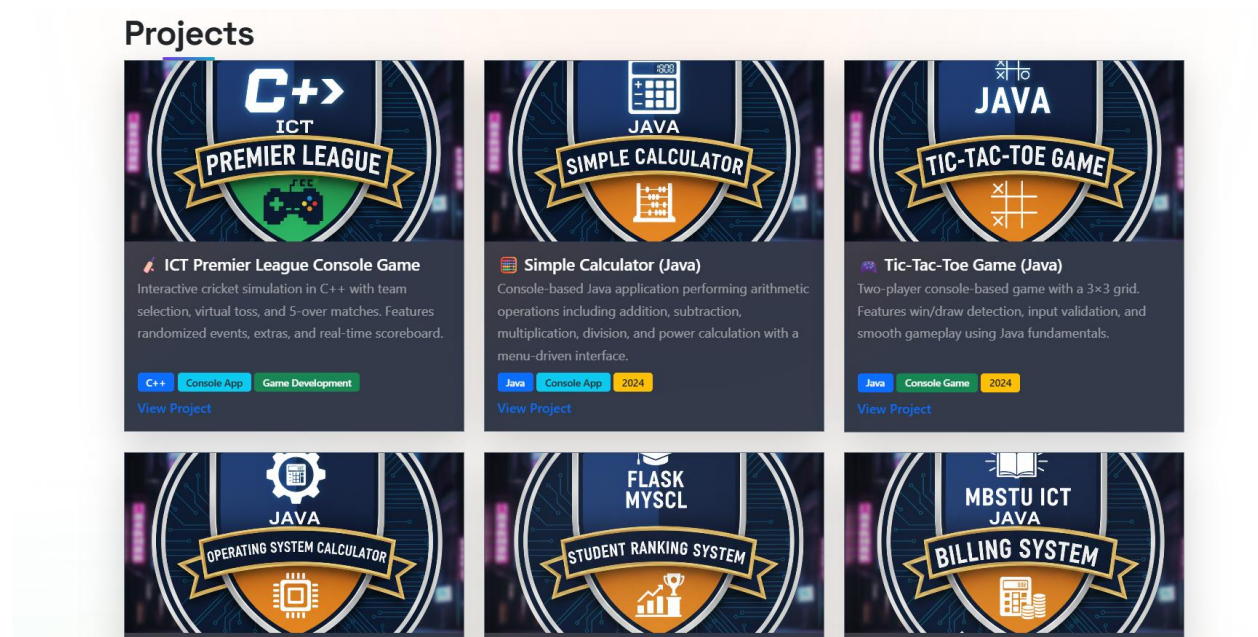


Fig 05: Projects Section

7. Developing the “Contact” Section:

Code:

```
<section class="contact-section my-5" id="contact">
  <div class="container">
    <h1 class="text-center section-title" data-aos="zoom-in" data-aos-offset="300" data-aos-duration="500">Contact</h1>
    <p class="text-center text-white-50 mb-5" data-aos="fade-up">Let's connect and create something amazing together</p>
  </div>

  <div class="container">
    <div class="row g-4 justify-content-center">
      <div class="col-lg-10" data-aos="fade-up">
        <div class="contact-info-card glass p-5 rounded-4">
          <div class="row g-4">
            <div class="col-md-6">
              <h3 class="text-white mb-4 fw-bold">Let's collaborate, learn, and build</h3>
              <p class="contact-text mb-5 fs-5">Reach out to discuss coding ideas, IoT experiments, or cybersecurity puzzles. I respond promptly.</p>
              <div class="contact-item mb-4" data-aos="fade-up" data-aos-delay="100">
                <div class="contact-icon-wrapper mb-3">
                  <i class="fa-solid fa-envelope contact-icon"></i>
                </div>
                <h5 class="text-white mb-2 fw-semibold">Email</h5>
                <a href="mailto:mugdha.saha67@gmail.com" class="contact-link contact-text d-block mb-1">mugdha.saha67@gmail.com</a>
                <a href="mailto:kuldip.saha67@gmail.com" class="contact-link contact-text d-block mb-1">kuldip.saha67@gmail.com</a>
                <a href="mailto:it22018@mbstu.ac.bd" class="contact-link contact-text d-block">it22018@mbstu.ac.bd</a>
              </div>
              <div class="contact-item mb-4" data-aos="fade-up" data-aos-delay="200">
                <div class="contact-icon-wrapper mb-3">
                  <i class="fa-solid fa-phone contact-icon"></i>
                </div>
                <h5 class="text-white mb-2 fw-semibold">Phone</h5>
                <p class="contact-text mb-1">+880 1711 123456</p>
                <p class="contact-text mb-1">+880 1711 234567</p>
              </div>
            </div>
          </div>
        </div>
      </div>
    </div>
  </div>
```

➤ Visual Overview of the “Contact” Section:

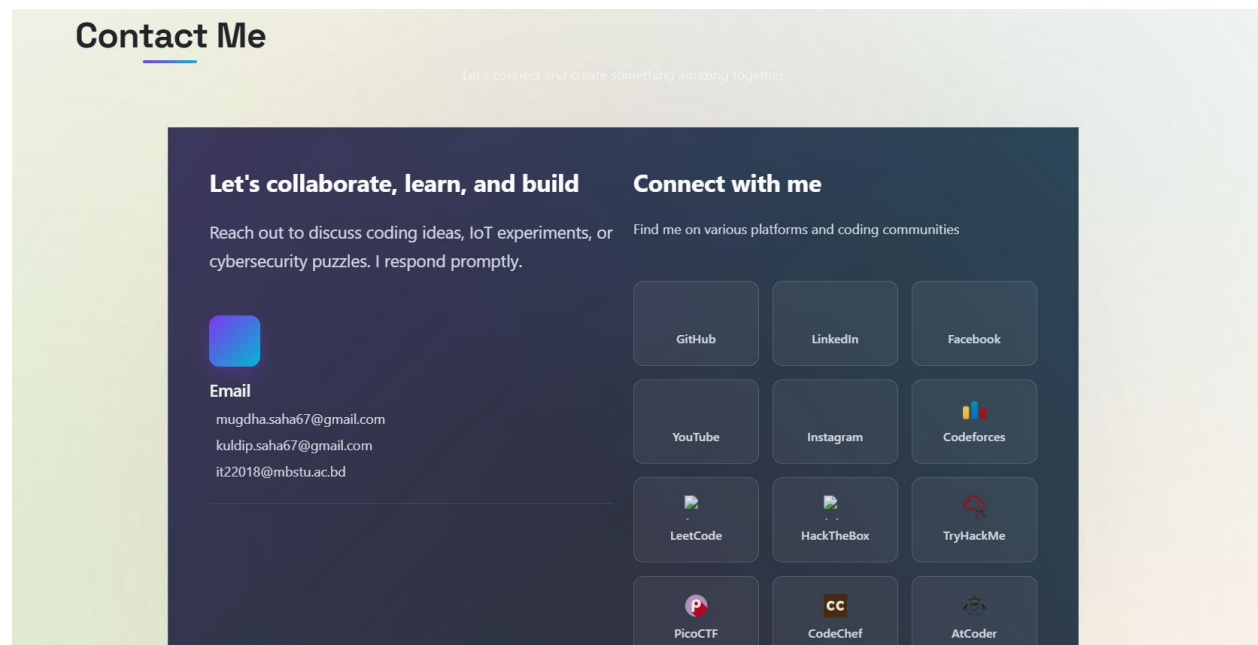


Fig 06: Contact Section

Live link of GitHub: [Mugdha Saha | Portfolio](#)

Conclusion:

This lab successfully demonstrated the development and deployment of a responsive personal portfolio website using HTML5 and Bootstrap 5. The integration of Google Fonts and AOS animations enhanced the visual appeal and interactivity of the website. The experiment helped in understanding responsive web design principles and the effective use of modern front-end frameworks.